
Transformations — Diagnosing a Wrong Centre, Vector or Scale Factor

Three pieces of data decide a transformation, and almost every wrong image comes from one of them being wrong rather than from bad arithmetic.

- **Vector.** A translation vector is *image minus pre-image*, in that order. Reverse the order and you get the translation that undoes it.
- **Scale factor.** A scale factor is a *ratio* of distances from the centre, never a difference.
- **Centre.** A dilation moves each point along the ray from *its own centre*. Multiplying the coordinates only works when that centre is the origin.
- Every claim here can be tested. Where a length is asked for, use

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

and give two decimal places when the answer is not exact.

1. **Warm-up.** A translation maps $A(-3, 2)$ to $A'(4, -1)$. Write the translation vector $\langle a, b \rangle$.

Answer: _____

2. Find and fix. Lee says the translation that maps $P(2, 5)$ to $P'(-3, 1)$ is $\langle 5, 4 \rangle$.

Lee subtracted in the wrong order. Explain what that does to a vector, then write the correct translation vector.

Answer: _____

3. Find and fix. Kim translates $A(1, 2)$ and $B(5, 5)$ by the vector $\langle 3, -1 \rangle$ and records the images as $A'(4, 1)$ and $B'(8, 5)$.

(a) Find AB . (b) Find $A'B'$ using Kim's coordinates, to two decimal places. (c) A translation is a rigid motion, so what do your two answers prove about Kim's work? Give the corrected y -coordinate of B' .

Answer: _____

4. Find and fix. A dilation centred at the origin maps $P(3, 4)$ to $P'(9, 12)$. Jo writes: “the scale factor is 6, because $9 - 3 = 6$.”

(a) Find the distance from the origin to P . (b) Find the distance from the origin to P' . (c) Use those two distances to find the real scale factor, and say in one sentence why a difference can never be a scale factor.

Answer: _____

5. Find and fix. A dilation has centre $C(2, 1)$ and scale factor 2. It maps $A(5, 3)$ to A' . Ravi doubles the coordinates of A and writes $A'(10, 6)$.

(a) Ravi used the wrong centre. Which centre did his method actually use? (b) Find the correct x -coordinate and y -coordinate of A' , using $A' = C + k(A - C)$. (c) Find CA' to two decimal places.

Answer: _____

6. Show what you know. Two dilations both use scale factor 2, but one is centred at C_1 and the other at C_2 , two different points.

(a) Sketch a triangle and both of its images on the grid below. Label the two centres and both images.

(b) Describe how the two images compare in *size* and in *position*, and say why.

(c) Suppose you are handed a point and its image under an unknown dilation. Explain how you would locate the centre, and why your method works.

